

# Homework 9

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## Problem 1.

*Proof.* Let  $(x_n)$  be a sequence of positive terms which converge to  $x > 0$ . Convergent sequences are bounded, so  $\exists M \in \mathbb{R} \ni x_n \leq M$ .  $0 < x_n < M \implies x_n = |x_n| \leq M$ .  $\forall \epsilon > 0$ . By the definition of limit,  $\exists K \in \mathbb{N} \ni \forall n \geq K, |x_n - x| < \frac{\epsilon}{|x| + |M|}$ .

$$\begin{aligned} |x_n^2 - x^2| &= |x_n^2 - xx_n + xx_n - x^2| \\ &= |(x_n - x)(x + x_n)| \\ &= |x_n - x| |x + x_n| \\ &\leq |x_n - x| (|x| + |x_n|) \text{ (triangle inequality)} \\ &\leq |x_n - x| (|x| + |M|) \text{ (since } |x_n| \leq M) \\ &< \frac{\epsilon}{|x| + |M|} (|x| + |M|) \\ &= \epsilon. \end{aligned}$$

Thus  $\lim x_n^2 = x^2$

□

## Problem 2.

*Proof.*  $\forall n \in \mathbb{N}, x_n = \frac{1}{\frac{1}{n} + 4} = \frac{n}{1 + 4n}$ .

$\forall \epsilon > 0$ . By Archimedean Property,  $\exists K \in \mathbb{N} \ni K > \frac{1}{16\epsilon} \implies \epsilon > \frac{1}{16K}$ .  $\forall n \geq K$

$$\begin{aligned} \left| \frac{n}{4n+1} - \frac{1}{4} \right| &= \left| \frac{4n - (4n+1)}{16n+4} \right| \\ &= \left| \frac{-1}{16n+4} \right| \\ &= \frac{1}{16n+4} \\ &< \frac{1}{16n} \text{ (since } 16n+4 > 16n) \\ &\leq \frac{1}{16K} \text{ (since } n \geq K \implies 16n \geq K) \\ &< \epsilon \end{aligned}$$

Thus,  $\lim x_n = \frac{1}{4}$ .

□

**Problem 3a.**

*Proof.* Suppose  $x_n = \frac{(-1)^n}{n^3}$ .

$\forall \epsilon > 0$ . By Archimedean property,  $\exists K \in \mathbb{N} \ni K \geq \frac{1}{\epsilon^{\frac{1}{3}}} \implies \epsilon > \frac{1}{K^3}$ .  $\forall n \geq K$ .

$$\begin{aligned} \left| \frac{(-1)^n}{n^3} \right| &= \frac{1}{n^3} \\ &\leq \frac{1}{K^3} \text{ (since } n \geq K \implies n^3 \geq K^3) \\ &< \epsilon \text{ (since } \epsilon > \frac{1}{K^3}) \end{aligned}$$

Thus,  $\lim x_n = 0$  □

**Problem 3b.**

*Proof.* Suppose  $x_n = (-1)^n(1 + \frac{1}{n})$ .

Take the subsequences,  $y_n = x_{2n} = (-1)^{2n}(1 + \frac{1}{2n}) = 1 + \frac{1}{2n}$  and  $z_n = x_{2n-1} = (-1)^{2n-1}(1 + \frac{1}{2n-1}) = -1 - \frac{1}{2n-1}$ .

Limit of constants are equal to the constant, thus  $\lim_{n \rightarrow \infty} 1 = 1$  and  $\lim_{n \rightarrow \infty} (-1) = -1$ .

$\forall \epsilon > 0$ . By Archimedean property,  $\exists K \in \mathbb{N} \ni K \geq \frac{1}{2\epsilon} + 0.5 \implies \epsilon > \frac{1}{2K-1}$ .  $\forall n \geq K$ .

$$\begin{aligned} \left| \frac{1}{2n-1} \right| &= \frac{1}{2n-1} \\ &\leq \frac{1}{2K-1} \text{ (since } n \geq K \implies 2n-1 \geq 2K-1) \\ &< \epsilon \end{aligned}$$

Thus  $\lim \frac{1}{2n-1} = 0$ .

By operation of limits theorem,  $\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} (1 + \frac{1}{2n}) = \lim_{n \rightarrow \infty} (1) + \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n} = 1$  and  $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} (-1 - \frac{1}{2n-1}) = \lim_{n \rightarrow \infty} (-1) - \lim_{n \rightarrow \infty} (\frac{1}{2n-1}) = -1 - 0 = -1$ .

Thus  $\lim y_n = 1$  and  $\lim z_n = -1$ . Thus subsequence of  $x_n$  have different limits, thus because of the divergence criteria given by subsequences,  $x_n$  is divergent. □

**Problem 3c.**

*Proof.* Suppose  $x_n = (-1)^n n^3$ .

$\forall r \in \mathbb{R}$ .

1. If  $r \leq 0$ .  $x_1 = 1 > r \implies r$  is not an upper bound for  $(x_n)$ .

2. If  $r > 0$ ,

- (a) If  $\lceil r \rceil$  is even, let  $K := \lceil r \rceil + 2$ , thus  $K$  even and  $K > \lceil r \rceil \geq r$ . Note  $K > 1$ . Thus  $x_K = (-1)^K K^3 = K^3 > K > r$ .
- (b) If  $\lceil r \rceil$  is odd, let  $K = \lceil r \rceil + 1$ , thus  $K$  even and  $K > \lceil r \rceil \geq r$ . Note  $K > 1$ . Thus  $x_n = (-1)^K K^3 = K^3 > K > r$ .

Thus  $r$  is not an upper bound for  $(x_n)$ .

Thus  $(x_n)$  is unbounded. By contrapositive of the convergence  $\implies$  bounded theorem,  $(x_n)$  is divergent.  $\square$

**Problem 3d.**

*Proof.* Suppose  $x_n = \frac{3n^3 - n + 4}{6n^3 + n + 5}$ .  $\forall \epsilon > 0$ , By the Archimedean property,  $\exists K \in \mathbb{N} \ni K > \frac{\sqrt{3}}{2\sqrt{\epsilon}} \implies \epsilon > \frac{3}{14K^2}$

$$\begin{aligned}
\left| \frac{3n^3 - n + 4}{6n^3 + n + 5} - \frac{1}{2} \right| &= \left| \frac{6n^3 - 2n + 8 - (6n^3 + n + 5)}{12n^3 + 2n + 10} \right| \\
&= \left| \frac{-3n + 3}{2(6n^3 + n + 5)} \right| \\
&= \left| \frac{-3}{2} * \frac{n - 1}{6n^3 + n + 5} \right| \\
&= \frac{3}{2} \left( \frac{n - 1}{6n^3 + n + 5} \right) \\
&< \frac{3}{2} \left( \frac{n}{6n^3 + n + 5} \right) \\
&< \frac{3}{2} \left( \frac{n}{6n^3 + n} \right) \\
&< \frac{3}{2} \left( \frac{1}{6n^2 + n^2} \right) \\
&< \frac{3}{2} \left( \frac{1}{7n^2} \right) \\
&\leq \frac{3}{14K^2} \\
&< \epsilon
\end{aligned}$$

Thus  $\lim x_n = \frac{1}{2}$ .  $\square$

**Problem 4a.**

*Proof.* (Base Case): For  $n = 0$ ,  $x_0 = 5 > -3$

(Inductive Hypothesis): Suppose the statement holds for  $n$  ie suppose  $x_n > -3$

(Inductive step):  $x_n > -3 \implies x_n/3 > -1 \implies x_n/3 - 2 = x_{n+1} > -1 - 2 = -3$ .

Thus by induction,  $x_n > -3$ .  $\square$

**Problem 4b.**

*Proof.* (Base Case): For  $n = 0$ ,  $x_0 = 5$  and  $x_1 = \frac{5}{3} - 2 = -\frac{1}{3} < x_0$ . Thus  $x_0 > x_1$

(Inductive Hypothesis): Suppose  $x_n < x_{n+1}$

(Inductive Step):  $x_n < x_{n+1} \implies x_n/3 < x_{n+1}/3 \implies x_n/3 - 2 = x_{n+1} < x_{n+1}/3 - 2 = x_{n+2}$ .

Thus by induction,  $(x_n)$  is monotonically decreasing.  $\square$

**Problem 4c.**

*Proof.* By monotonic convergence theorem, since  $(x_n)$  has a lower bound and is decreasing,  $(x_n)$  is convergent.  $\square$

**Problem 4d.**

*Proof.* Let  $L := \lim x_n$ .

$\lim x_{n+1} = L$  since limit of a subsequence is equal to the limit of the sequence.

By the operation of limits,  $\lim x_{n+1} = \frac{1}{3} \lim x_n + \lim(-2)$ . Limit of a constant term, is the constant, thus  $\lim x_{n+1} = \frac{1}{3}L - 2$ .

By Uniqueness of Limits,  $\frac{1}{3}L - 2 = L \implies -2 = \frac{2}{3}L \implies L = -3$ .  $\square$

**Problem 5a.**

*Proof.*  $\forall \alpha \in \mathbb{R}$ . By the definition of proper divergence,  $\exists K \in \mathbb{N} \ni \forall n \geq K, x_n > \alpha \implies -x_n < -\alpha$ , since multiplication by a negative real number flips the ordering sign. Thus  $-x_n < -\alpha$ .

By the definition of proper divergence,  $(-x_n)$  diverges to  $-\infty$ .  $\square$

**Problem 5b.**

*Proof.* Let  $\alpha = 1$ . For this  $\alpha$ ,  $\exists K \ni \forall n \geq K, x_n > \alpha$ .  $\forall m, n \in \mathbb{N} \ni m > n \geq K$ .

$$\begin{aligned} \left| \sum_{k=1}^m x_k - \sum_{k=1}^n x_k \right| &= \left| \sum_{k=n+1}^m x_k \right| \\ &= \sum_{k=n+1}^m x_k \quad (\text{since } x_k > \alpha > 0 \quad \forall k \in \mathbb{N} \ni n < k \leq m) \\ &> (m - n)(\text{since } x_k > 1) \\ &> 1 \end{aligned}$$

Thus by definition of cauchy,  $(\sum x_n)$  is not cauchy. Since convergence  $\iff$  cauchy,  $(\sum x_n)$  is not diverges.  $\square$